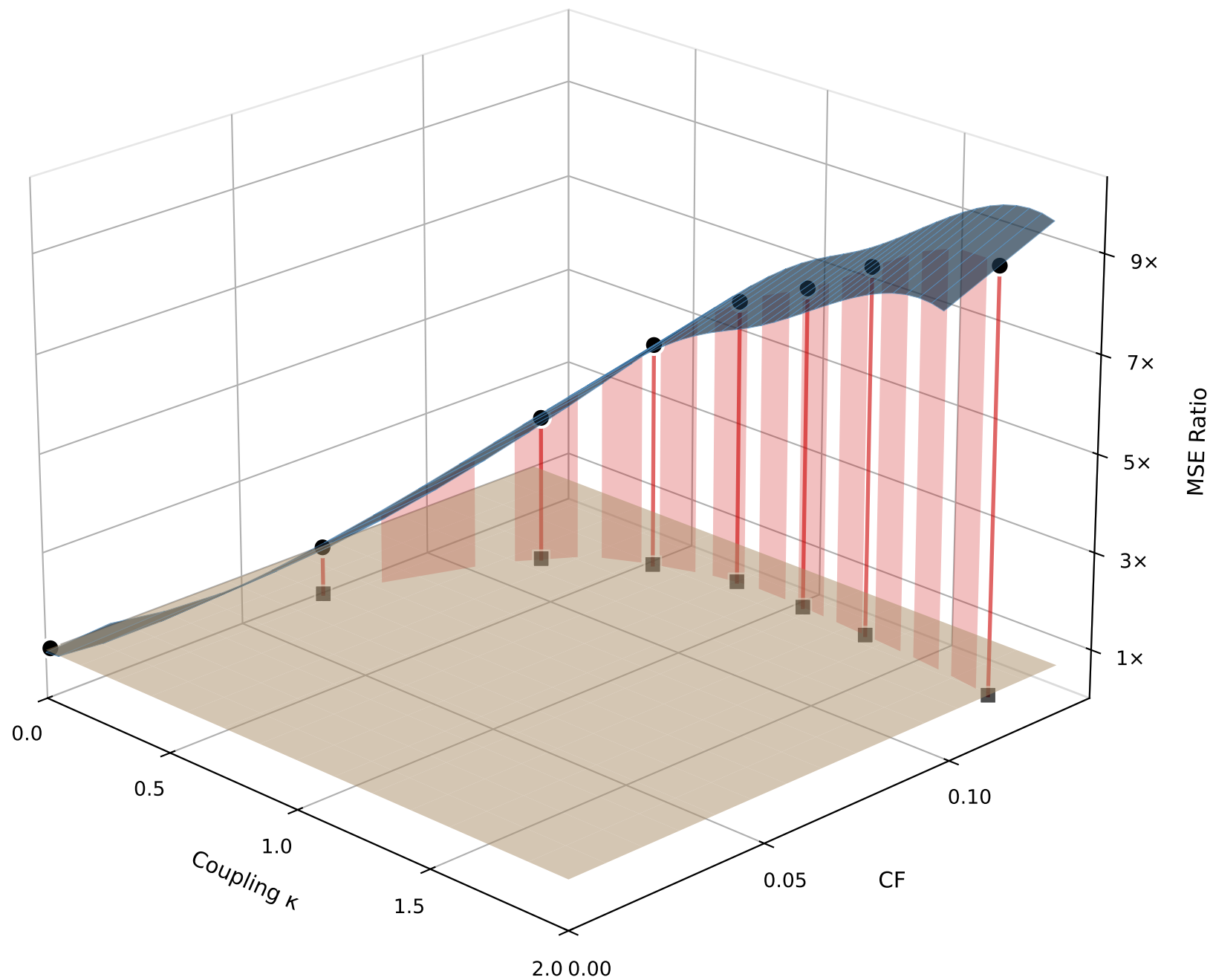


Information-Geometric View: MFVI as KL Projection

SVF simulation data ($N = 8,000$)



$\mathcal{M}$ Full Manifold

Curved surface where true posterior $p(z, x|y)$ lives

$\mathcal{M}_F$ Mean-Field Plane

Flat surface at MSE = 1.0x
Factorized $q(z)q(x)$

Lost Fidelity

Shaded volume between $\mathcal{M}$ and $\mathcal{M}_F$

Measurements

- On manifold (true)
- On plane (projected)

$\kappa=0.5$: Δ MSE = 3.0x
 $\kappa=1.0$: Δ MSE = 5.8x
 $\kappa=2.0$: Δ MSE = 8.5x